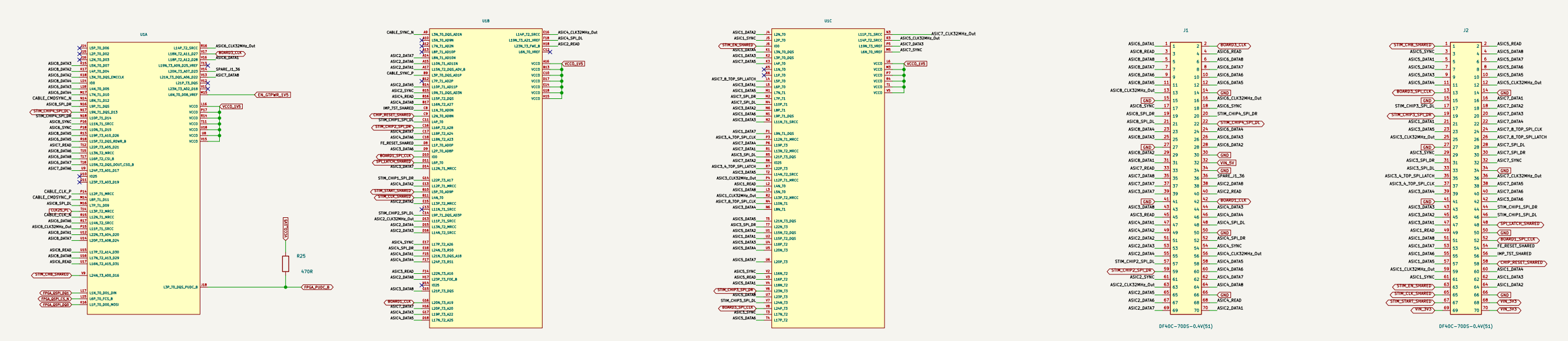
01. FPGA I/O + ASIC INTERFACES + 25 MHz CLOCK | Trenz p5



1. Function: U1 / J1 / J2: Route 121 FPGA lines (120 signals for eight 1.5 V ASICs and one spare, J1.36) through two 70–contact Hirose DF40C–70DS mezzanines with 15 grounds (J1 14/15/27/30/34/41/50/66, J2 14/15/27/34/41/50/66). J2.68–70 carry VIN_3V3 and J1.32 carries VIN_5V (unused provision, 0.3 A, between grounds J1.30 and J1.34) to the FPGA–LDO board. Contact order follows pin map V6: top chips ASIC3/4/7/8 have their own SPI data, clock and latch; each returned 32 MHz clock is on a P–side MRCC/SRCC ball.

2. Source / changes: Codex custom ASIC interface. V8 FPGA–side pin remap (Howard approved 2 Nov 2026): ASIC groups moved to the bank nearest their connector, each returned clock on a clock–capable pin. V11 (Howard approved 2 Nov 2026): J1/J2 DF40C–60DS to DF40C–70DS (Hirose DF40 catalog Jun 2024 p.8 and p.14 Note 3, guideline ETAD–H1015–00 sec 1.2.1, https://www.hirose.com/en/product/series/DF40); the V11 reading that corner contacts are not energized is superseded: Gerald (7 Nov 2026) confirmed DF40C corner contacts 1/2/69/70 are live; the fittings are separate MP pads. ASIC_REFERENCE.md sections 2–5; Chip_FPGA Interface.pptx slides 13–17/25–28; Pins.xlsx. Trenz TE0714 REV04 p5 I/O bank use adapted. https://www.trenz-electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH-TE0714-04-42I-7-L.PDF V30 (Howard 4 Nov 2026): R25 0402 –> 0201 Yageo RC0201 1 %, value unchanged.

V43 (Howard 9 Nov 2026, option (a) 'Re–pin in the FPGA', Claude): contact order = FPGA–LDO chat remap V3 (2026–11–09_FPGA_Work/03_FPGA_LDO_J1_J4_Plan_and_Remap_V3/remap_table_V3.csv; J2 = V2) so the LDO board routes J1–>J4 and J2–>J3 as ordered buses; U1 ball assignment re–solved on the 8L V1 lane plan (all 8 groups 11/11 in their clock's bank: bank 14 ASIC1 P15, ASIC8 T14; bank 15 ASIC2 D13, ASIC3 E13, ASIC4 E16; bank 34 ASIC5 R2, ASIC6 P4, ASIC7 R3; AMD xc7a35tcsg325 package file); 103 U1 nets reassigned, 14 balls freed (no–connect). BOARD2/4_CLK, BOARD2/4_SPI_CLK and NONSTIM_BOARD1/2_SPI_DL/DR removed: the PCB–5 ASIC board leaves J3/J4 pins 2/4/6/8/24/58 open, so stacked boards share the near board's clock/SPI/STIM lines. STIM_EN/START/CHB/CLK_SHARED now on J1.59/63/57/61 (FPGA–LDO 10k idle pull–downs move with them). J1.69/J1.70 added to VIN_3V3 (LDO board about 155–215 mA plus start–up vs 0.3 A per contact; all of it still comes from J3.19 VIN_5V through U16). Unverified: Gerald/Howard to confirm identical configuration of stacked boards; XDC/gateware pin constraints not yet written; ASIC timing at the new lane lengths (longest ASIC1 CLK32 J1.53 –> P15, about 12 mm); mating LDO V26 schematic.

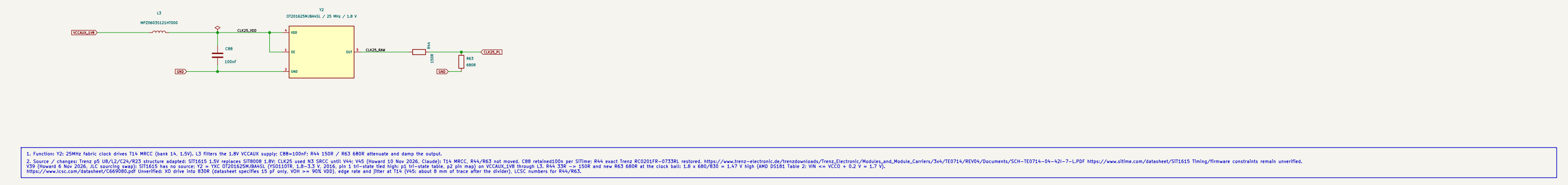
V44 (Howard 9 Nov 2026, "Yes, study re–pin"): ASIC6_DATA6 moves from P6 to J6 (both bank 34, ASIC6 group stays 11/11 in its clock bank); its planned lane passed 0.06 mm from the JTAG TDI T9 dog–bone via. J2.59 and every other J1/J2 contact are unchanged. Codex re–solve, unverified in Vivado.

V45 (Howard 10 Nov 2026, "B: both on J2" and "Yes, T14"; Claude): edge–connector layout for the 8L V3 board (J1 east edge, J2 west edge). Contact order = FPGA–LDO chat pin map V5 (2026–11–10_FPGA_Work/15_FPGA_LDO_Pin_Map_V5_and_Schematic_V30/J1_J2_pin_map_V5.csv); this supersedes the V43 remap V3 order and the V43/V44 ball list. U1 balls re–solved by minimum total ball–to–contact distance on the 8L V3 placement: bank 34 ASIC1 N3, ASIC3 P4, ASIC5 R2, ASIC7 R3; bank 15 ASIC2 D13, ASIC4 E16; bank 14 ASIC6 R16, ASIC8 P15. CLK25_PL moves N3 –> T14 (IO_L13P_T2_MRCC_14) to free the fourth bank 34 clock ball. Nine J2 slow nets use bank 15 balls A10/B10/B11/C8/C9/D8/D9/D10/D11. VIN_3V3 moves from J1.68–70 to J2.68–70 (still three contacts). Unverified: no lane routing tried yet; ASIC timing at the new lane lengths (longest straight line 14 mm slow, 11.5 mm ASIC1); XDC/gateware constraints not written; mating FPGA–LDO schematic V30; clock placement in Vivado.

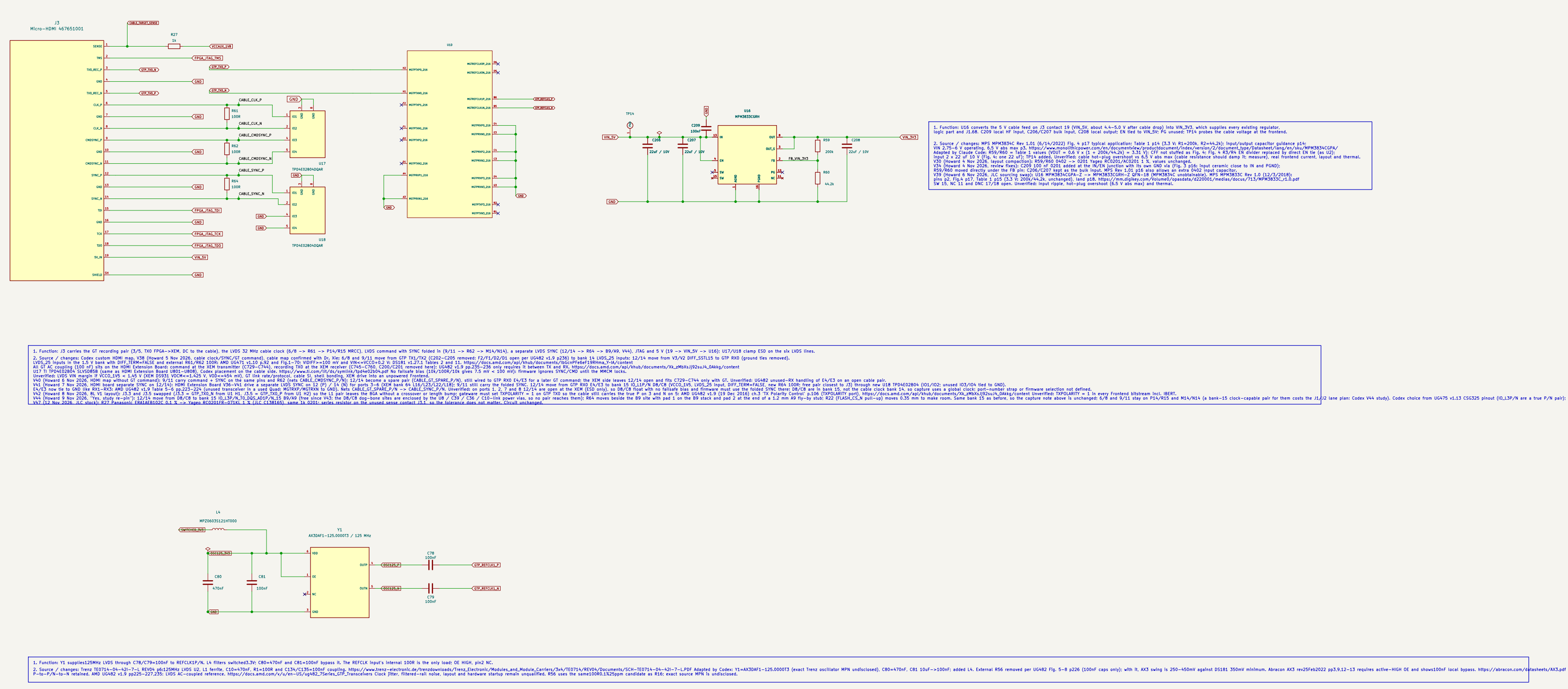
V46 (Howard 10 Nov 2026, "Yes, re–pick first" and "finish it up"; Claude): J2 FPGA balls re–picked so that every J2 lane is an ordered bus on the proposed 12–layer stack; 53 U1 balls change, the J1 balls and every J1/J2 contact are unchanged (pin map V5 frozen by Howard). Clock balls: ASIC1 R2, ASIC3 P4, ASIC5 R3, ASIC7 N3 (P–side MRCC/SRCC, bank 34; the four J2 groups stay whole in bank 34). Slow nets: BOARD3_SPI_CLK V8, STIM_CHIP3_SPI_DL/DR V7/V6, STIM_EN J6 (bank 34); STIM_CHB V9 (bank 14); STIM_CHIP1_SPI_DR/DL G14/C11, SPI_LATCH D11, BOARD1_SPI_CLK D10, FE_RESET D8, IMP_TST C8, CHIP_RESET C9, STIM_CLK B11, STIM_START B10 (bank 15). Method: integer program over ball, lane layer and channel slot (L10 for the east pad row, L8/L6 for the west pad row and J2.60/62/64), then native KiCad copper on a scratch trial copy: J2 58/58 and J1 50/50 connected ball to contact, DRC 0 errors. Unverified: the 12–layer stack and its via rules are not confirmed by JLC; eight power via stacks were moved in the trial and no planes exist yet; ASIC timing (longest ASIC lane 14.0 mm = ASIC1 clock; group spreads 4.2–6.7 mm); XDC not run in Vivado; the V45 statement about nine bank 15 slow balls is superseded by this list.

V47 (Howard 12 Nov 2026, after the meeting with the ASIC designer: pin order final, top chips keep their own SPI clock and latch, 5 V provision; "move forward"; Claude): nets per pin map V6 (2026–11–12_FPGA_Work/09_Pin_Map_V6/PIN_MAP_V6.csv; published as the working map, not frozen). 12 new inputs on former spare contacts: J1.29/31/33/35/37/39/43/45/47/49 and J2.40/42 to balls K17/U12/T12/V13/H16/D14/G15/F14/F15/E13 and U9/D9. 18 contacts renamed on their existing wire; 12 of them are now FPGA outputs (ASIC3/4/7/8_SPI_DL/DR, ASIC3_4_TOP_SPI_CLK/LATCH, ASIC7_8_TOP_SPI_CLK/LATCH). Two lane swaps keep the ASIC3 and ASIC7 clocks on clock balls: J2.25 ASIC3_CLK32MHz_Out R7 –> P4, J2.35 ASIC3_4_TOP_SPI_LATCH P4 –> R7, J2.36 ASIC7_CLK32MHz_Out L4 –> N3, J2.26 ASIC7_8_TOP_SPI_LATCH N3 –> L4. J1.32 = VIN_5V (one contact, 0.3 A: Hirose DF40 catalog p.5, https://www.hirose.com/en/product/series/DF40), J1.30 = GND guard, J1.36 = SPARE_J1_36 on V14. DISC_OUT is not connected.

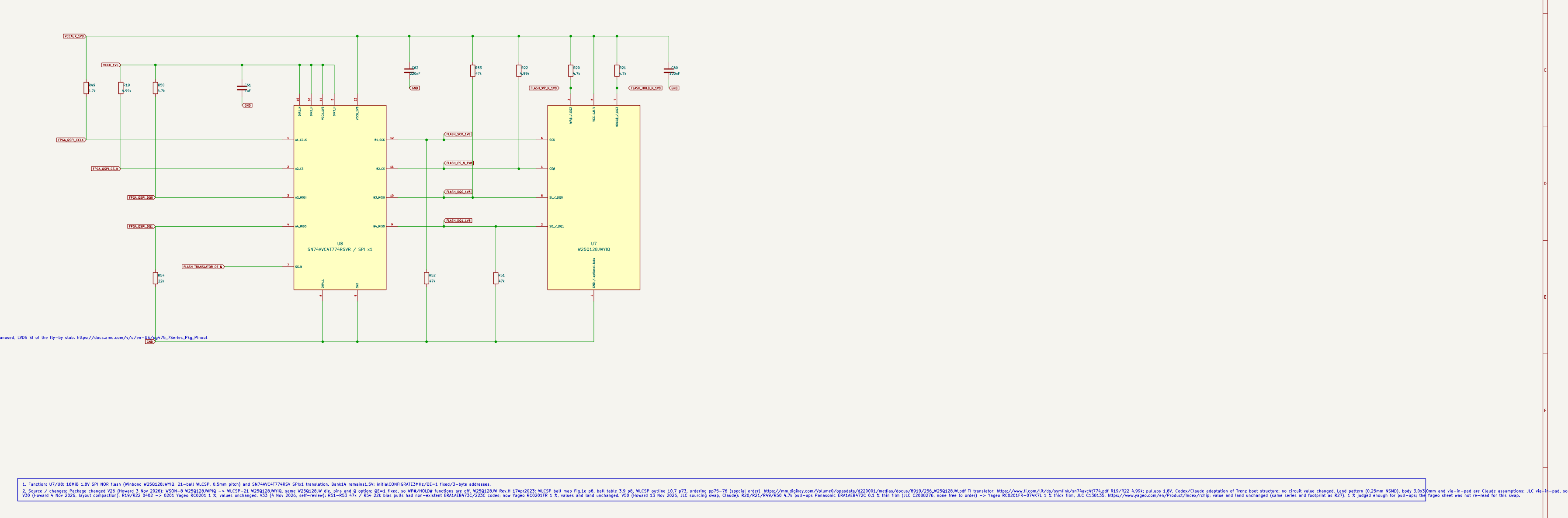
Ball functions: AMD xc7a35tcsg325pkg.txt, https://download.amd.com/adaptive-socs-and-fpgas/developer/adaptive-socs-and-fpgas/package-pinout-files/a7packages/xc7a35tcsg325pkg.txt Unverified: pin file not run in Vivado; ASIC3 and ASIC7 inputs span two or three banks (capture on a global clock); second functions of balls V13, U9, D9 read from pin names only; ASIC data timing still open; the pull–downs for the top SPI latches are on FPGA–LDO A V13; mating FPGA–LDO A V13.



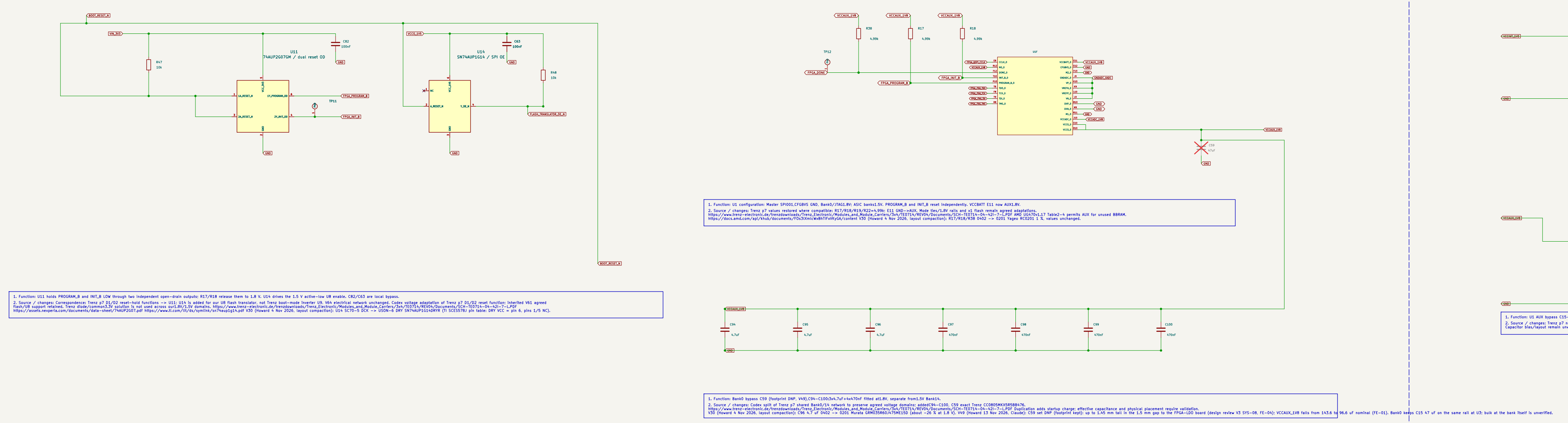
02. GTP CHANNELS + 125 MHz REFERENCE | Trenz p6



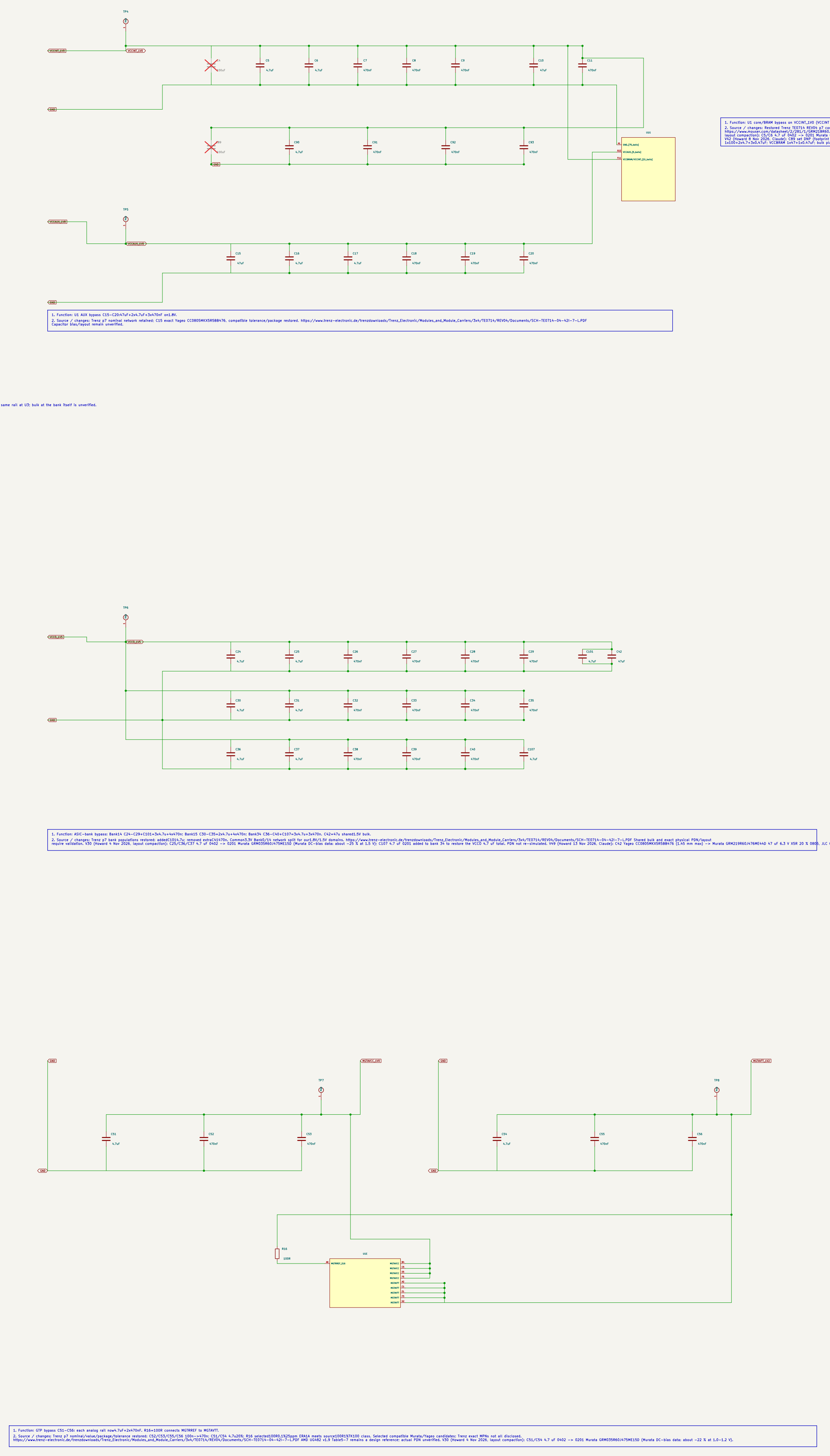
03. FLASH + SPI TRANSLATION | Trenz p6 adaptation



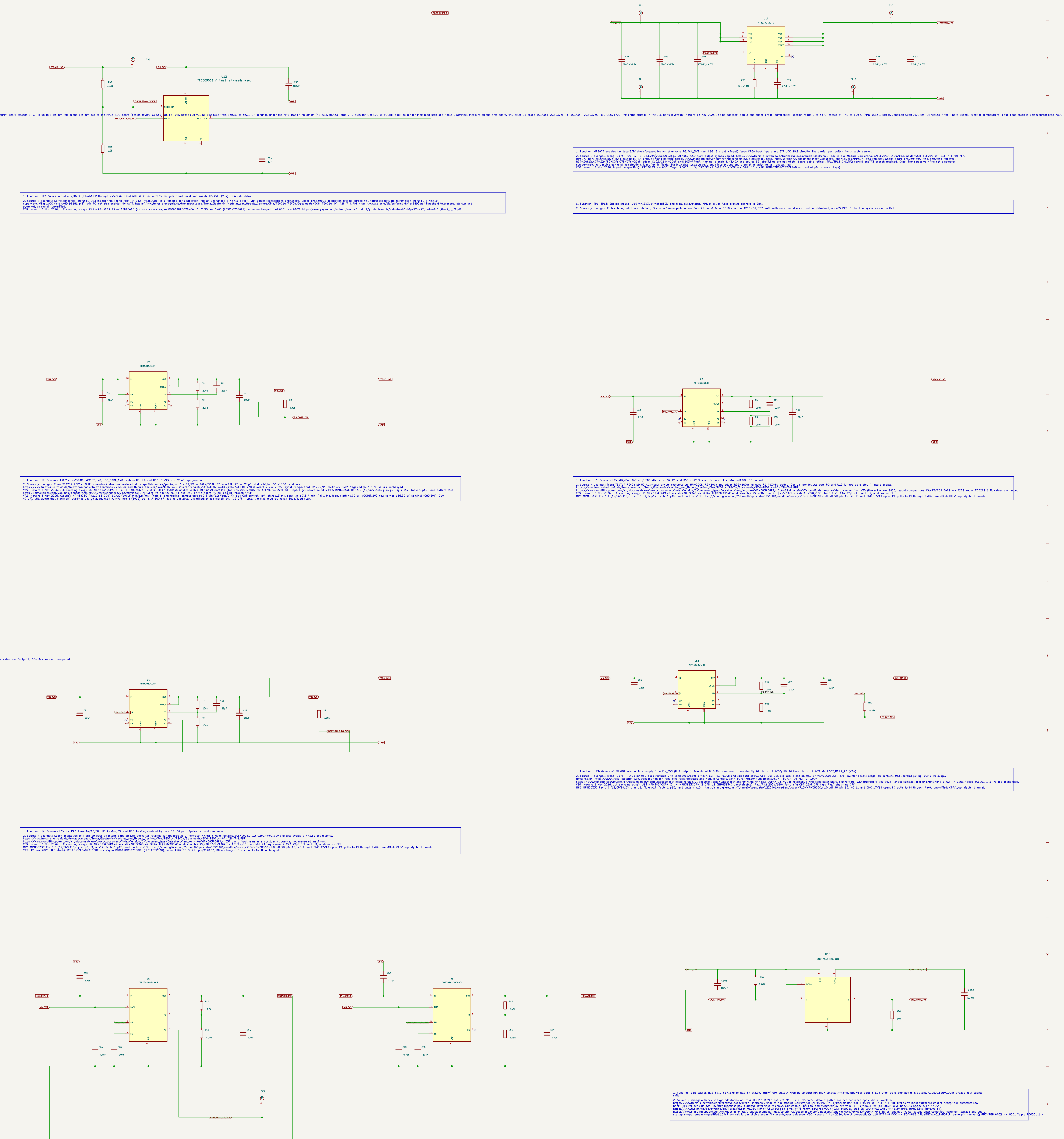
04. CONFIGURATION + RESET OUTPUTS | Trenz p7 roles



05. FPGA SUPPLY PINS + BYPASS + XADC | Trenz p7



06. REGULATORS + READINESS SUPERVISOR | Trenz p9 roles



Revision history (moved here from the title block in V48, 13 Nov 2026, so that the title fits its box; wording unchanged):
Artix Frontend V47 – nets per pin map V6 (12 new top–chip inputs, top–chip SPI outputs, 2 clock lane swaps, VIN_5V on J1.32, spare J1.36), R7/R27 = JLC BOM parts
V46 J2 balls re–picked in lane order (12–layer lane trial: J2 58/58, J1 50/50)
V45 edge connectors: J1/J2 pin map V5, U1 re–pin, CLK25 N3–>T14 (8L V3)
V44 cable pairs re–pinned near J3 (separate SYNC 12/14 to B9/A9, ASIC6_DATA6 P6–>J6)
V43 J1/J2 remap (FPGA re–pin, LDO V3 pin order)
V42 J3 TX0 pins 3/5 swapped (TXPOLARITY=1)
V41 separate LVDS SYNC on HDMI 12/14 to D8/C8 (HDMI board V41 map) + V39 sourcing swaps (schematic for the re–route)
V47 comment 1: HDMI: 3/5 GT TX0 recording; 6/8 LVDS 32 MHz clock; 9/11 LVDS command+SYNC; 12/14 LVDS SYNC (ports 3–6) to B9/A9 (V44)
V47 comment 2: V47 schematic from V46 (12 Nov 2026, Howard: move forward, finish this up); for 8L V3 on 14 layers V4; not fabrication–ready
V48 (13 Nov 2026, finishing pass; Claude): title block text only. No symbol, wire, label or field changed; netlist identical to V47.

Not fabrication–ready. Revision history: note above this box
V50 = V49 with R20, R21, R49, R50 as a JLC–stocked 1 % part; nets unchanged; for 14 layers V7
HDMI: 3/5 GT TX0; 6/8 LVDS clock; 9/11 command+SYNC; 12/14 SYNC (ports 3–6)

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Title: Artix Frontend V50 – nets per pin map V6

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